



Databricks Apps

Build secure data applications
with ease



Data apps were hard to build and productionize

Complex

Apps are bottlenecked around specialized resources, which slows the pace of development and makes them hard to maintain over time.

Time-consuming

Apps take longer to deploy due to increased reviews and iterations, often delaying the app's time to market. Infrastructure is extremely complex to deploy fast.

Vulnerable

Governing apps separately from underlying data is cumbersome and error-prone, resulting in poorly governed and insecure apps.

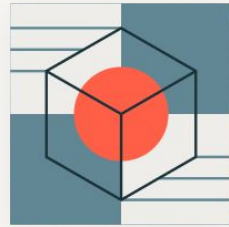
Introducing Databricks Apps

Quickly Build Secure Data Applications



Simple

Simplified app development in the technologies developers already know makes it easy to build instead of being bogged down in learning other technologies



Production-Ready

IT has high confidence in the production readiness of the apps developers are building and these can be deployed much faster, and get to market quickly

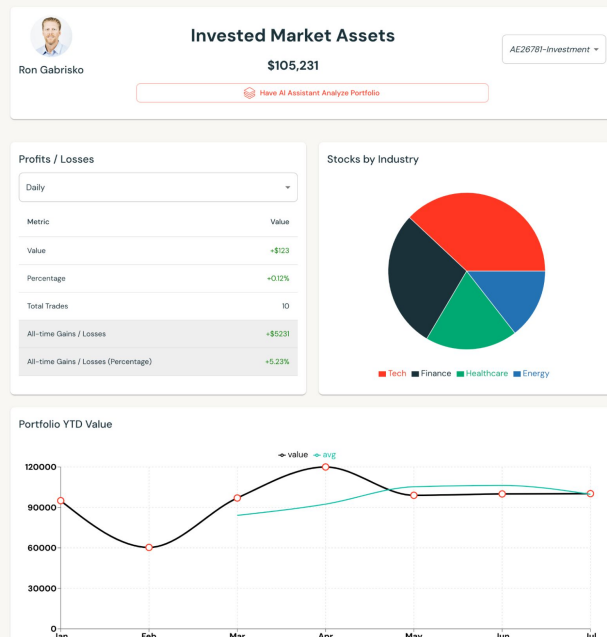


Built-in Governance

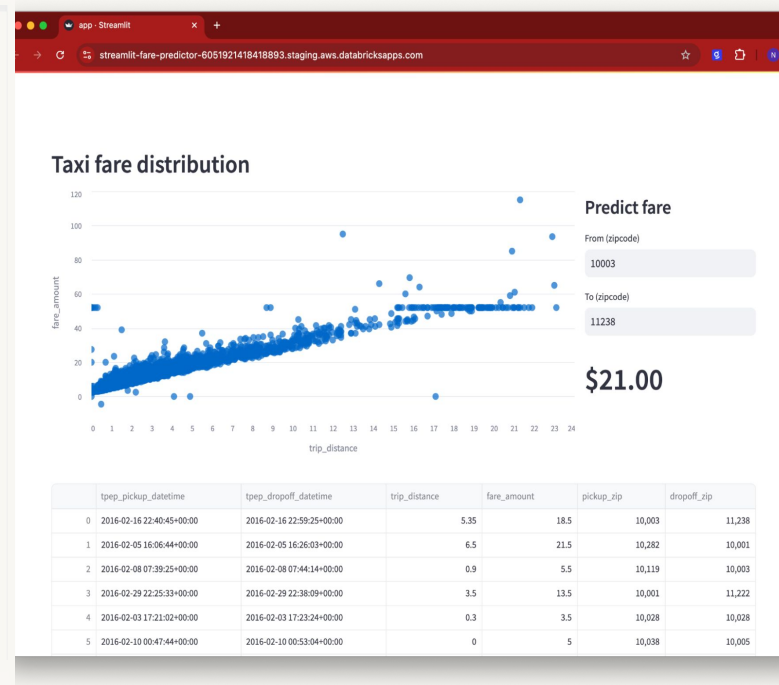
Data applications are inherently secure and well-governed, leading to greater data security

Use Databricks Apps if you want to build...

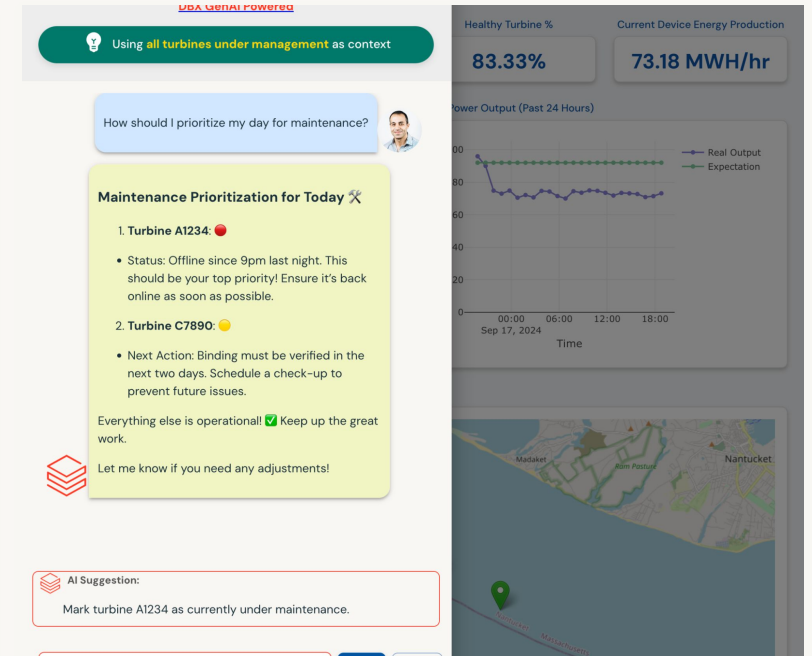
Interactive data apps

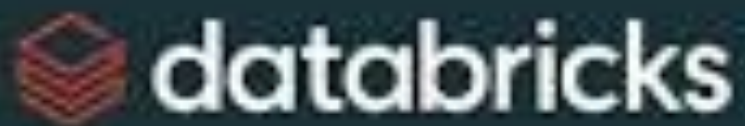


Predictive analytics



GenAI applications





Build Databricks Apps in 5 Mins!



Jackie Zhang
Technical Marketing - Databricks

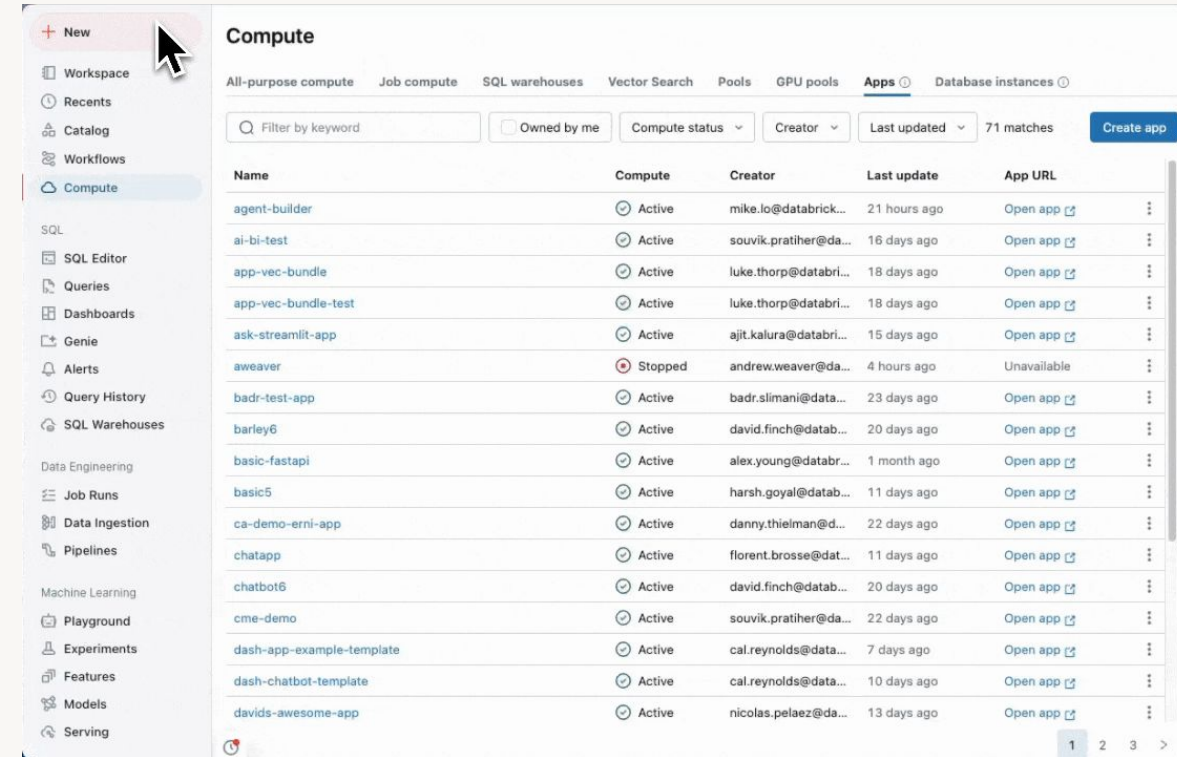


Simple

Use familiar Python frameworks such as Dash, Gradio and Streamlit, Shiny

Run directly within your Databricks environment, ensuring seamless access to data and AI models

Discover internal apps quickly using search and filtering in the app directory

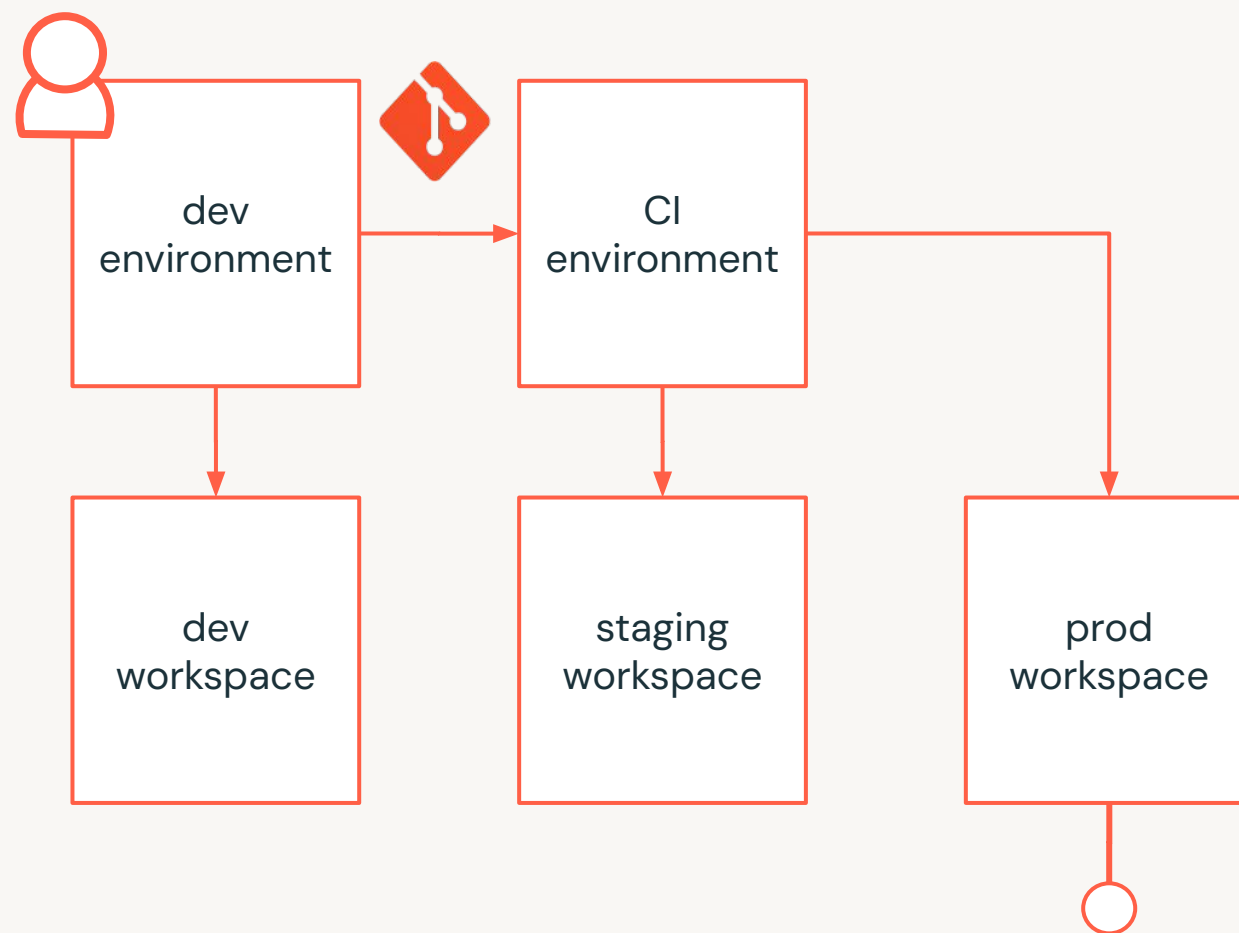


Production-ready

Apps run on automatically provisioned serverless compute, allowing deployment with ease

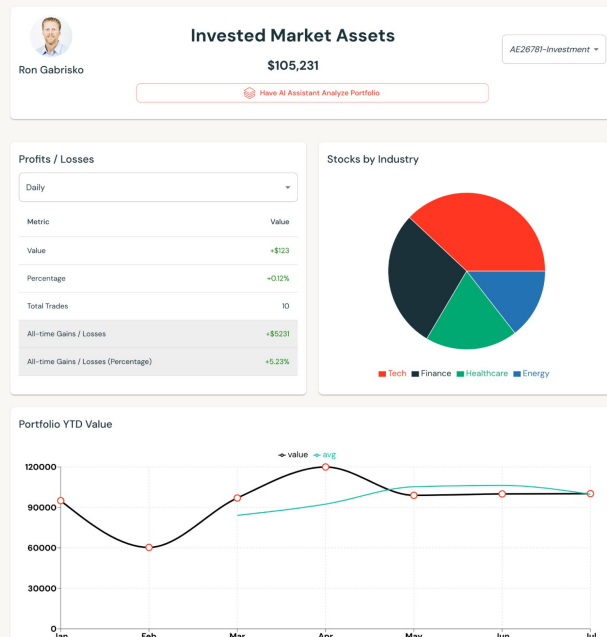
Built-in support for Git version control and CI/CD pipelines

Unique, predictable URL generated from deployment that developers can easily share with intended users

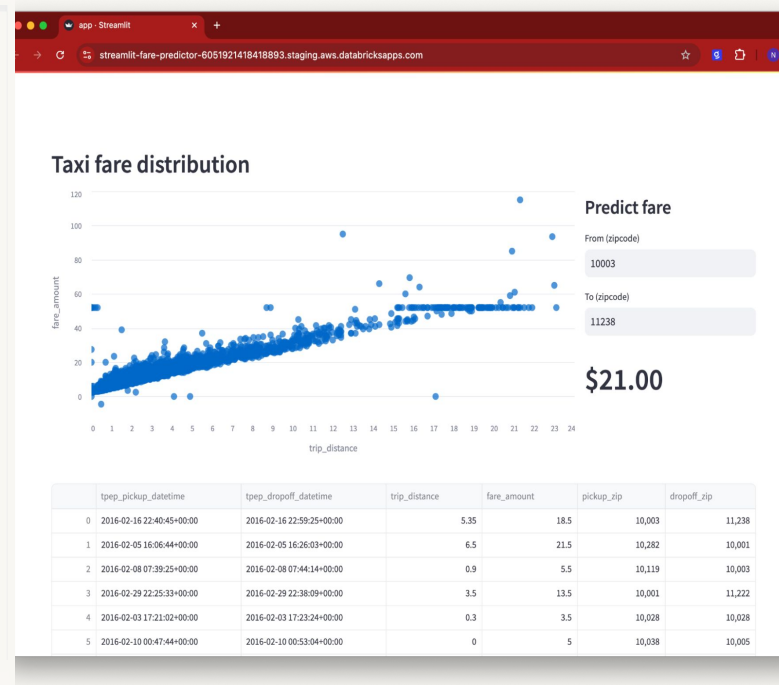


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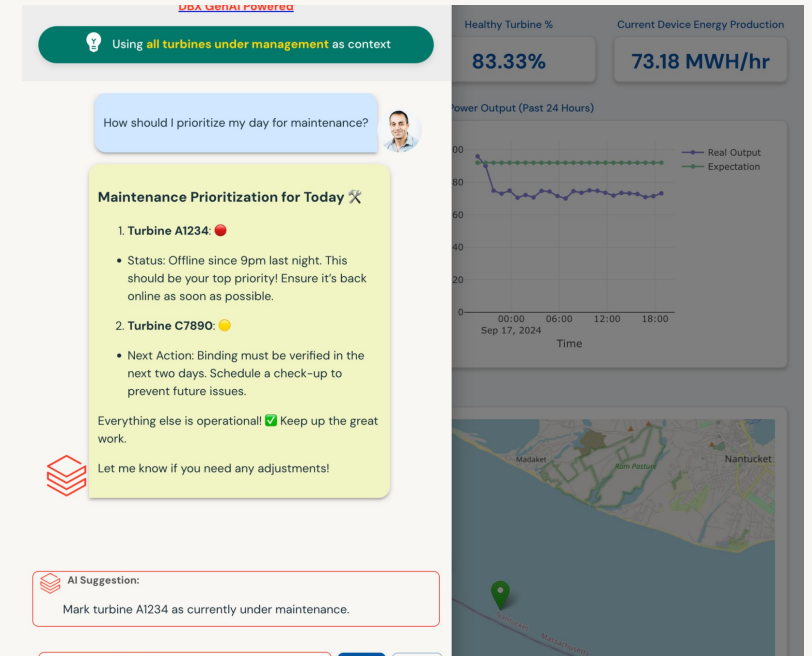
Interactive data apps



Predictive analytics



GenAI applications



Sharing apps outside the workspace

App users don't need access to a workspace

Permission Settings for:
app1

NAME	PERMISSION	
Admins	Can Manage	inherited
Data Dan (data.dan@databricks.com)	Can Manage	☐

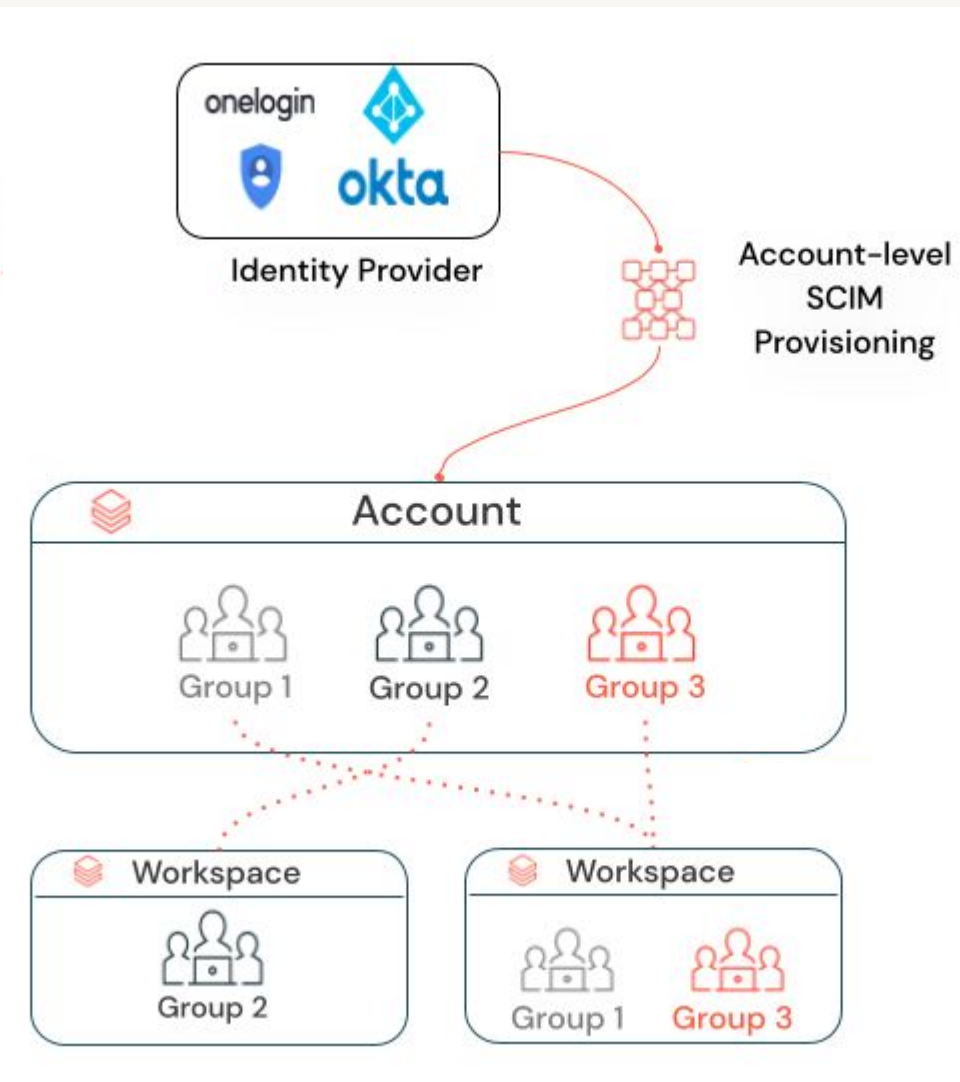
Select user, group or service principal... Can Manage ☐ + Add

Organization permissions

☒ Anyone in my organization can use

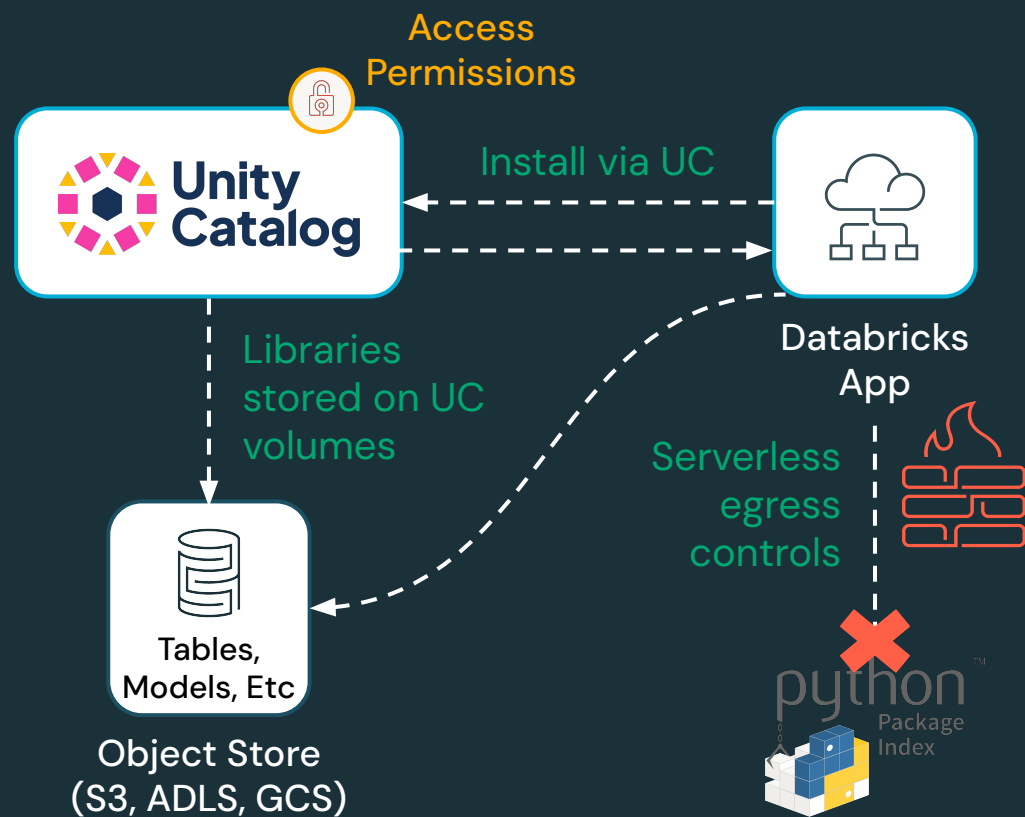
Cancel Save

- Can also name specific users
- App users need to be in your Databricks account, but not in a workspace
- Use SCIM sync to bring users from your identity provider into your Databricks account



Library Installation

Your Databricks Workspace



Libraries are installed at launch

- Libraries defined in requirements.txt are installed directly in the sandboxed container, running on your dedicated compute

Use public / cloud repositories or UC

- By default, libraries are downloaded from public repos like PyPi
- You can specify self-hosted repos (use Databricks secrets for credentials)
- You can also load libraries from your UC Volumes

Shared responsibility model

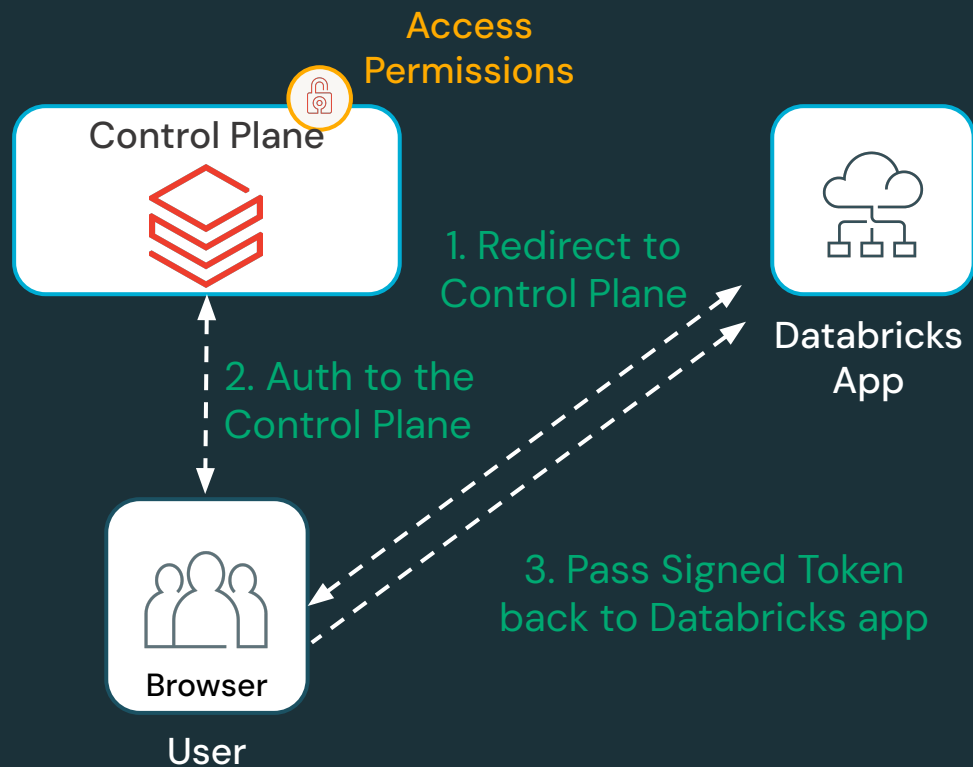
- Customers are responsible for the management and patching of these dependencies.

Serverless Egress Controls

- Block access to public repositories with serverless egress controls

Apps Authentication

Your Databricks Workspace



User Authentication

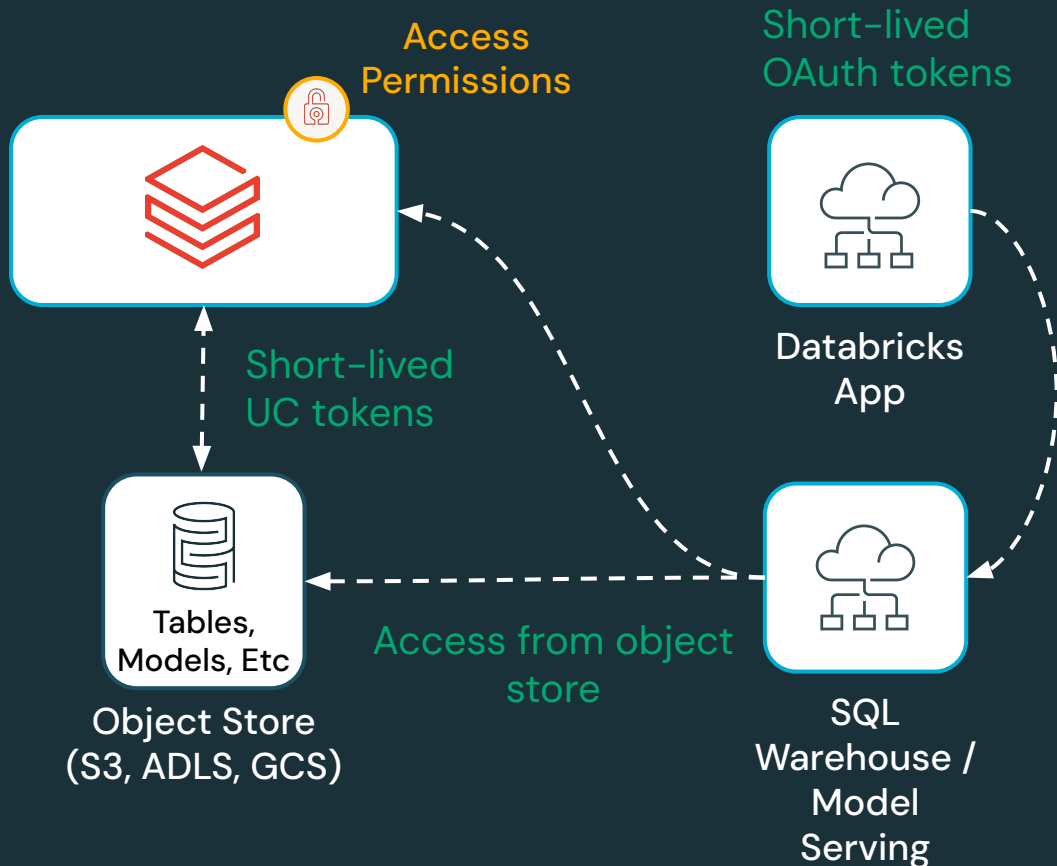
- Users connect to databricksapps.com URL
- An unauthenticated user gets an OAuth redirect to the control plane
- The control plane authenticates the user and directs them back to the app with the signed OAuth Scope

SSO, IP Access Lists Respected

- SSO + MFA are enforced as per the account/workspace/IdP configuration
- IP Access Lists and/or Frontend Private Link are respected as normal

Access to resources

Your Databricks Workspace



Dedicated service principal

- A dedicated service principal (SP) is automatically created for each app
- Access to tables, clusters, and more can be added for your SP, just like any other user or SP
- When you add a resource you can manage to your app, your SP will automatically be granted access.

On-behalf-of (OBO) user apps

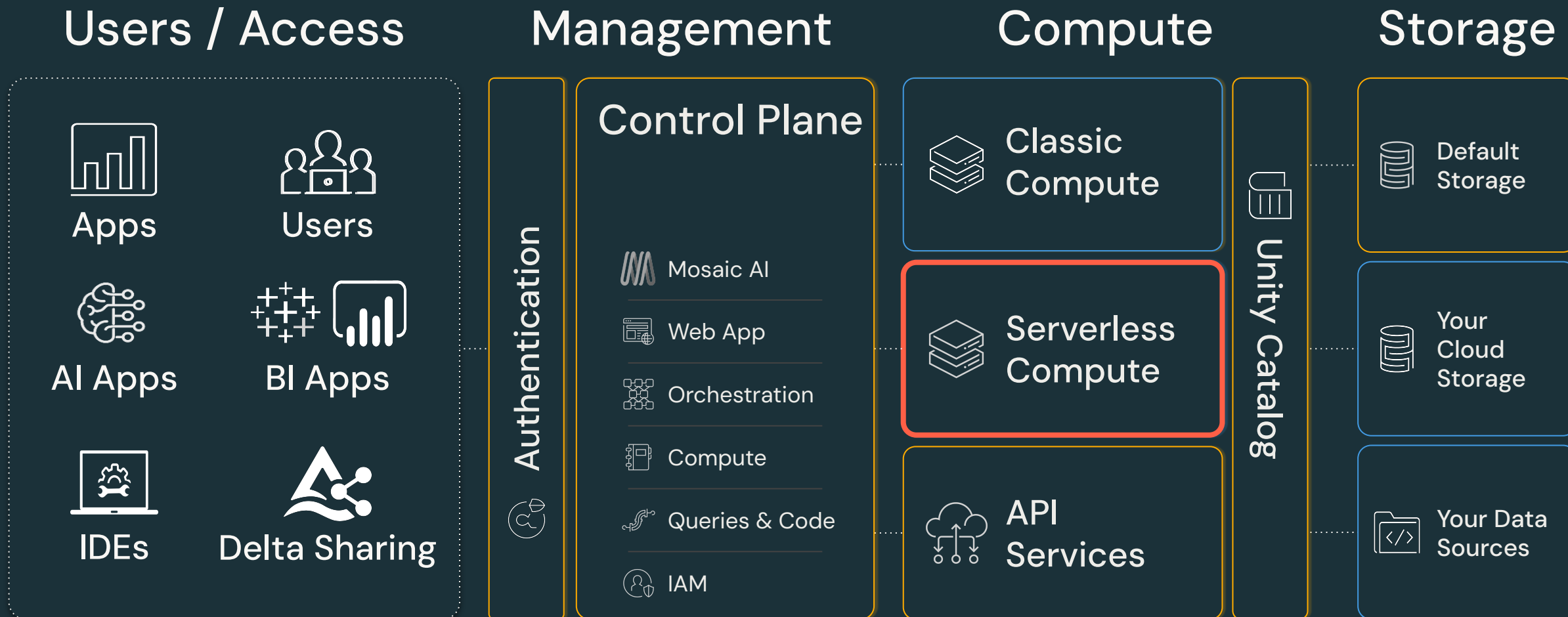
- Admins can optionally allow app authors to create OBO user apps
- OBO user apps leverage the logged in user's identity and permissions to access resources.

Best practice for compute

- Connect to SQL warehouses, jobs, or model serving directly from your app

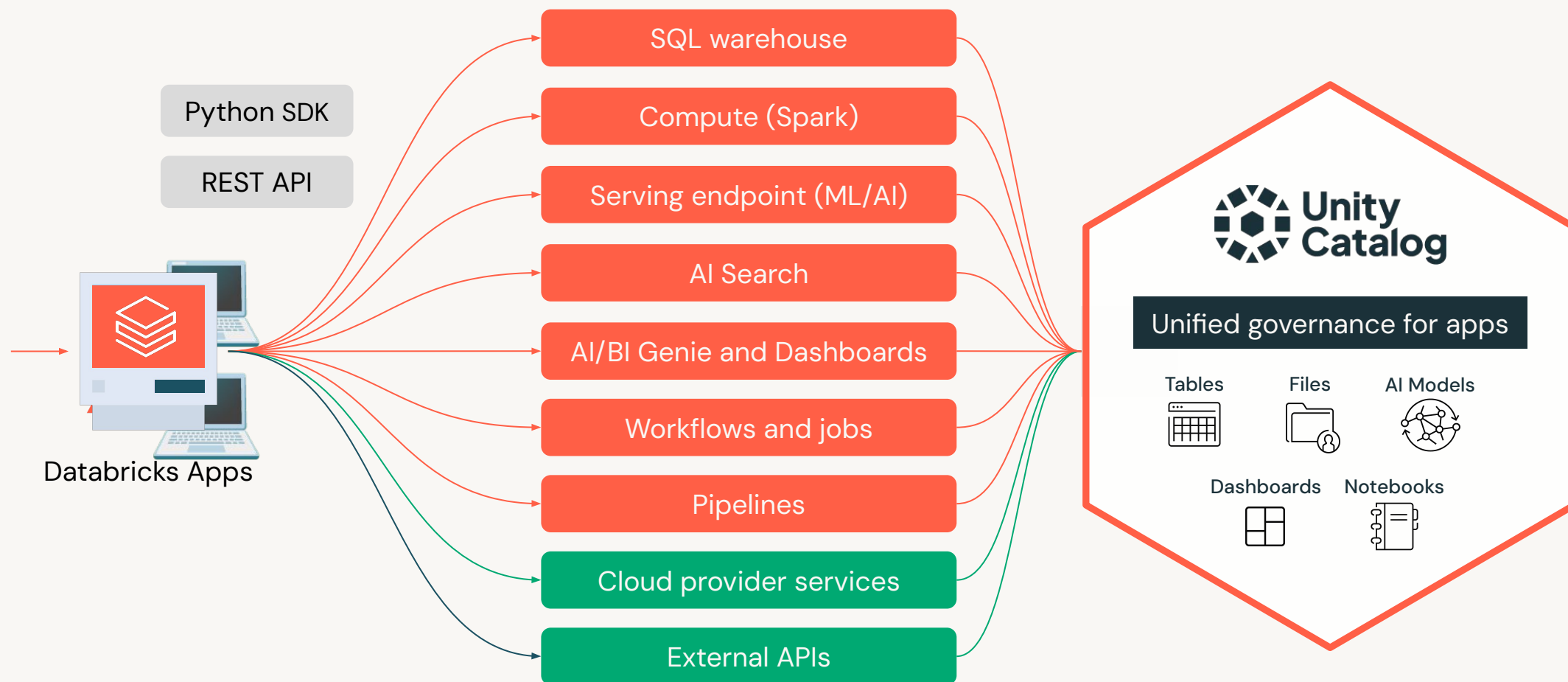
Databricks Architecture

Databricks Apps run in the Serverless Compute Environment



Apps integrate with your data and AI stack

Connect seamlessly with internal and external services



Databricks Apps Cookbook

Ready-to-use code snippets for building data and AI applications

- 10+ recipes for common Apps use cases
- Interactive demos, code snippets, requirements
- Deploy to Databricks or run locally
- Contributions welcome

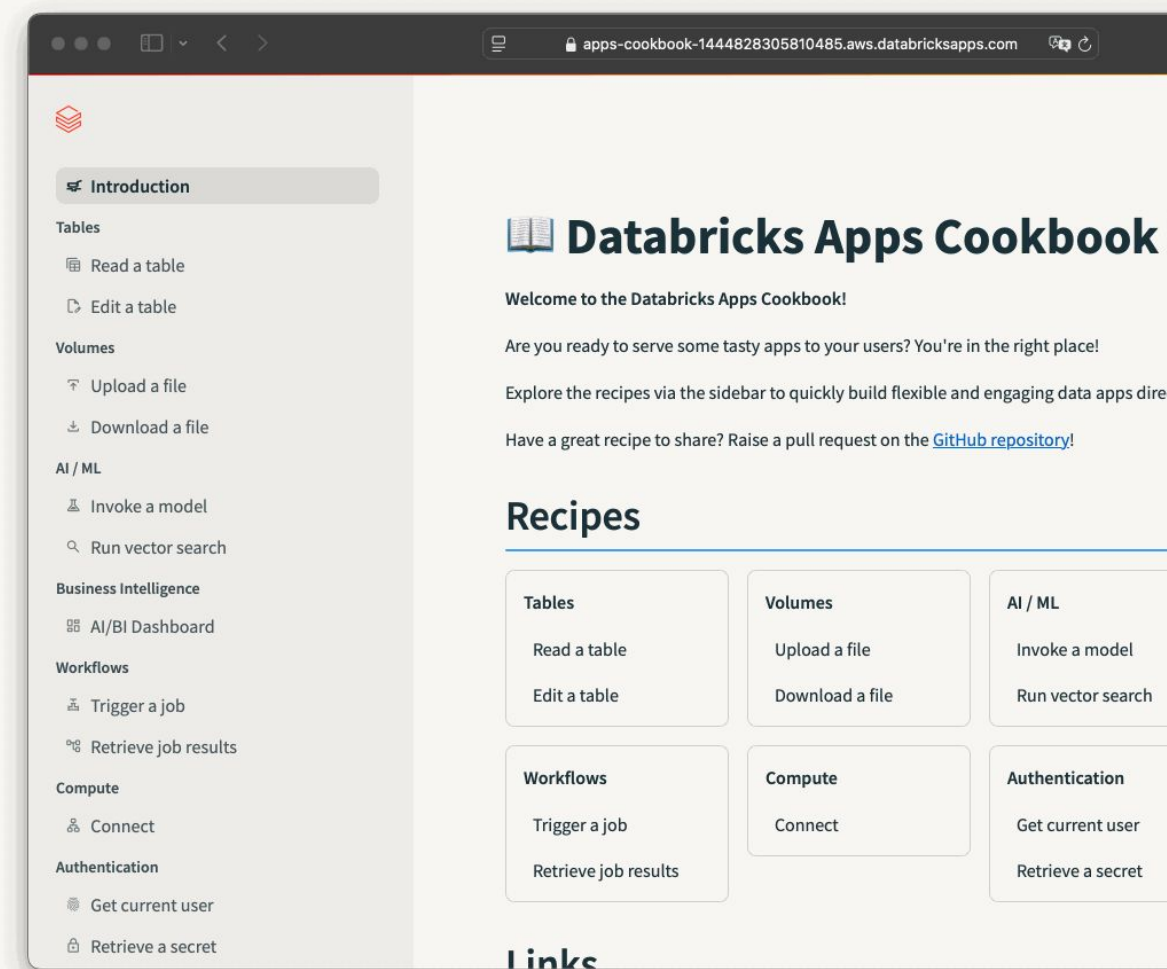
apps-cookbook.dev



Dash



Streamlit



Secret management

Use Databricks secret management to access external services

Automatically grant access to the secret

- If user deploying the app has permission to grant access
- Can also manually assign permissions to the service principal

Create new app

App name ⓘ
Name can't be changed once app is created

Description ⓘ

Advanced settings

^

App resources

+ Add resource

Resource type	Scope	Secret key
Secret	nick-scope ▾	password ▾
Permission Can read ▾		
Resource key secret		

Remove resource

Back

Create app

Accessing Platform Resources in the Code

1. Add an entry for the resource in the `app.yaml` configuration file using the SQL warehouse resource key. Because this is referencing the source of the parameter value and not the actual value, use `sql-warehouse` instead of `value`.

YAML

```
env:  
  - name: "DATABRICKS_WAREHOUSE_ID"  
    valueFrom: "sql-warehouse"
```

2. To reference the resource in your app code, use the value of the name field (`DATABRICKS_WAREHOUSE_ID` in this example) to refer to the configured key value.

Python

```
import os  
os.getenv('DATABRICKS_WAREHOUSE_ID')
```

Resources

Simplified access grants

Automatically grant access to the specified resources

- If user deploying the app has permission to grant access
- Can also manually assigned permissions to the service principal

Resources make it easier to move apps to new environments

Edit app

App name ⓘ
Name can't be changed once app is created

nick1

Description ⓘ

Advanced settings ^

App resources
+ Add resource

Resource type	Resource mapping
SQL warehouse	sqlw with long long long long long long long long I ▾
Permission	Resource key
Can use ▾	sql-warehouse

Remove resource

Back

Save

Templates

Also available as GitHub repo

Dash **Flask** **Gradio** **Streamlit**

Chatbot
Create a chat UI using a large language model and Model Serving.
Dependencies: Serving endpoint

Data app
Create an app that reads from a SQL warehouse and visualizes the data.
Dependencies: SQL warehouse

Hello world
Create a simple app.

Dash **Flask** **Gradio** **Streamlit**

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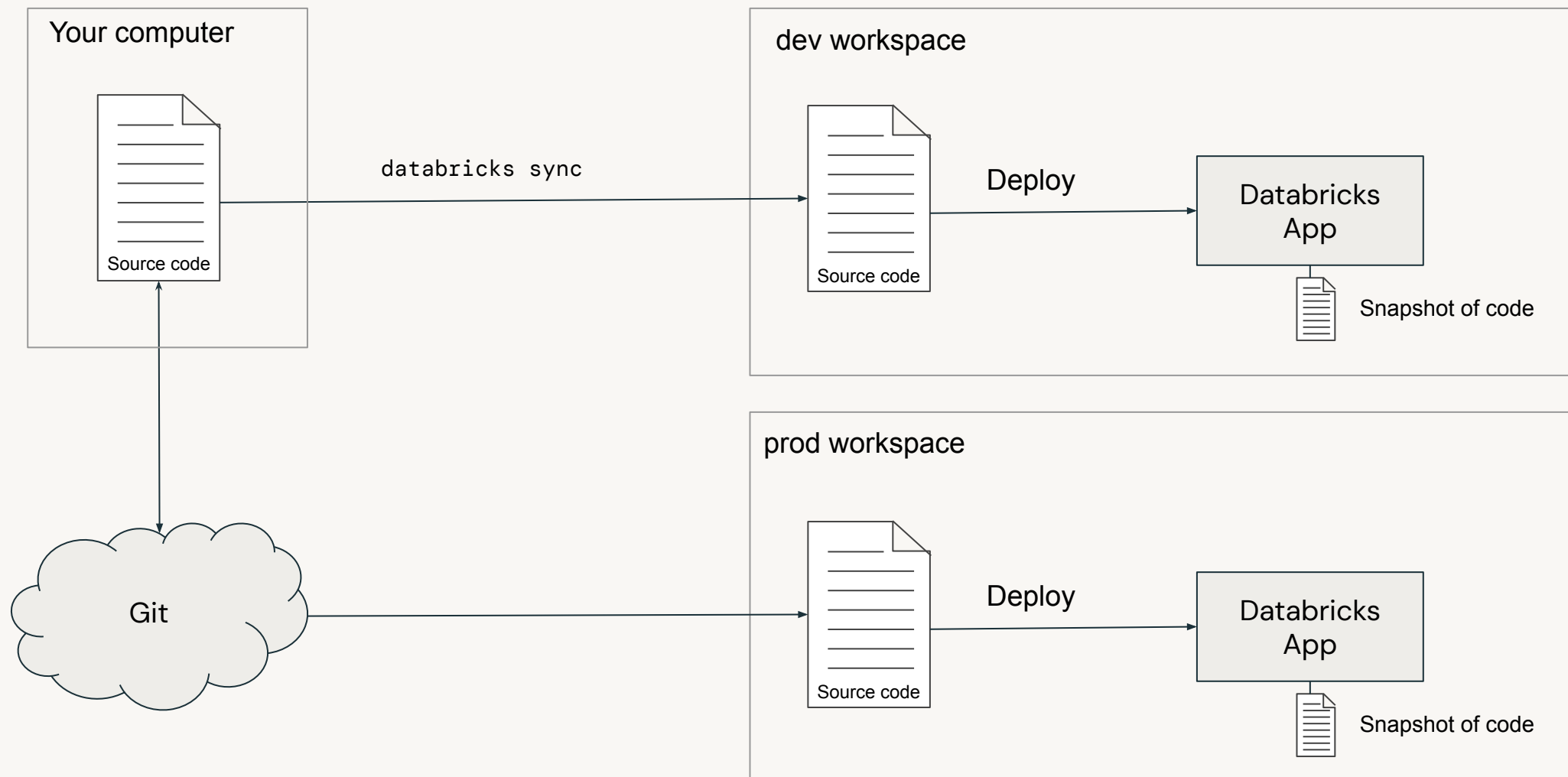
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Hello world
Create a simple app.

Typical Code Flow



Logs

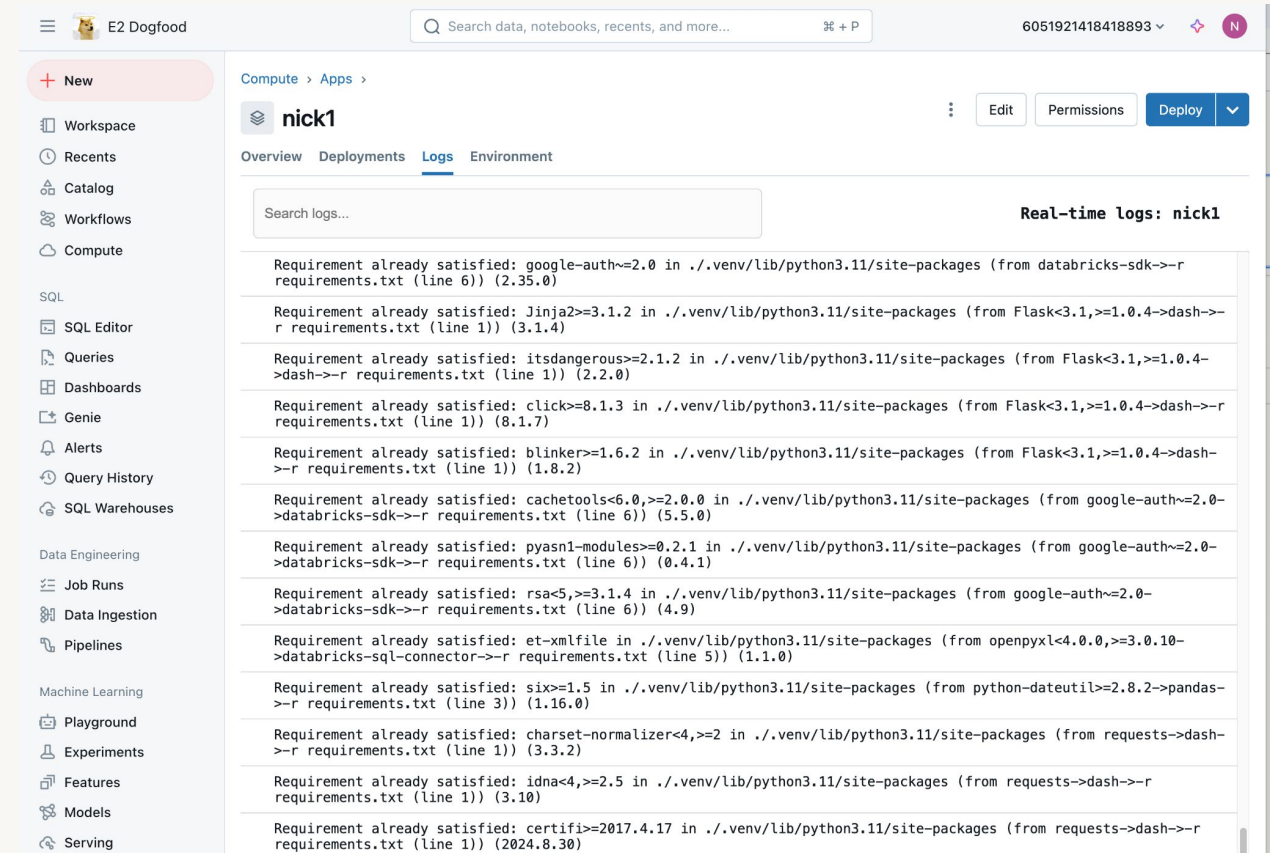
From both pip and the app

Logs of standard out and standard error

Erased when compute is stopped

Secrets are redacted

See also Deployments tab



The screenshot displays the Databricks interface for an application named 'nick1'. The left sidebar contains navigation options: New, Workspace, Recents, Catalog, Workflows, Compute, SQL, SQL Editor, Queries, Dashboards, Genie, Alerts, Query History, SQL Warehouses, Data Engineering, Job Runs, Data Ingestion, Pipelines, Machine Learning, Playground, Experiments, Features, Models, and Serving. The main panel shows the 'Logs' tab for the application 'nick1'. It includes a search bar for logs and a list of log entries under the heading 'Real-time logs: nick1'. The log entries are as follows:

- Requirement already satisfied: google-auth~=2.0 in ./venv/lib/python3.11/site-packages (from databricks-sdk->r requirements.txt (line 6)) (2.35.0)
- Requirement already satisfied: Jinja2>=3.1.2 in ./venv/lib/python3.11/site-packages (from Flask<3.1,>=1.0.4->dash->r requirements.txt (line 1)) (3.1.4)
- Requirement already satisfied: itsdangerous>=2.1.2 in ./venv/lib/python3.11/site-packages (from Flask<3.1,>=1.0.4->dash->r requirements.txt (line 1)) (2.2.0)
- Requirement already satisfied: click>=8.1.3 in ./venv/lib/python3.11/site-packages (from Flask<3.1,>=1.0.4->dash->r requirements.txt (line 1)) (8.1.7)
- Requirement already satisfied: blinker>=1.6.2 in ./venv/lib/python3.11/site-packages (from Flask<3.1,>=1.0.4->dash->r requirements.txt (line 1)) (1.8.2)
- Requirement already satisfied: cachetools<6.0,>=2.0.0 in ./venv/lib/python3.11/site-packages (from google-auth~=2.0->databricks-sdk->r requirements.txt (line 6)) (5.5.0)
- Requirement already satisfied: pyasn1-modules>=0.2.1 in ./venv/lib/python3.11/site-packages (from google-auth~=2.0->databricks-sdk->r requirements.txt (line 6)) (0.4.1)
- Requirement already satisfied: rsa<5,>=3.1.4 in ./venv/lib/python3.11/site-packages (from google-auth~=2.0->databricks-sdk->r requirements.txt (line 6)) (4.9)
- Requirement already satisfied: et-xmlfile in ./venv/lib/python3.11/site-packages (from openpyxl<4.0.0,>=3.0.10->databricks-sql-connector->r requirements.txt (line 5)) (1.1.0)
- Requirement already satisfied: six>=1.5 in ./venv/lib/python3.11/site-packages (from python-dateutil>=2.8.2->pandas->r requirements.txt (line 3)) (1.16.0)
- Requirement already satisfied: charset-normalizer<4,>=2 in ./venv/lib/python3.11/site-packages (from requests->dash->r requirements.txt (line 1)) (3.3.2)
- Requirement already satisfied: idna<4,>=2.5 in ./venv/lib/python3.11/site-packages (from requests->dash->r requirements.txt (line 1)) (3.10)
- Requirement already satisfied: certifi>=2017.4.17 in ./venv/lib/python3.11/site-packages (from requests->dash->r requirements.txt (line 1)) (2024.8.30)

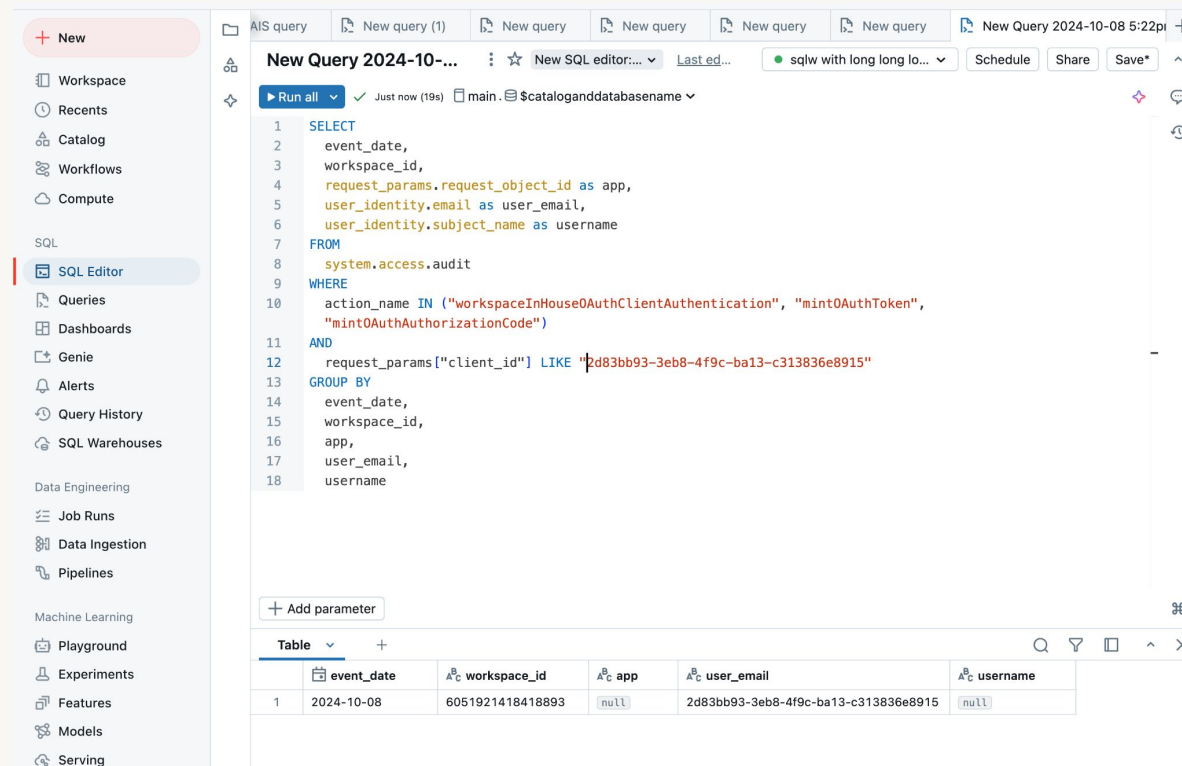
Audit logging

FROM `system.access.audit`

Logins to an app [[docs](#)]

Permission or sharing changes [[docs](#)]

[Audit log system table reference - Azure Databricks | Microsoft Learn](#)



The screenshot displays the Databricks SQL Editor interface. On the left is a sidebar with navigation options: New, Workspace, Recents, Catalog, Workflows, Compute, SQL (selected), SQL Editor, Queries, Dashboards, Genie, Alerts, Query History, SQL Warehouses, Data Engineering (Job Runs, Data Ingestion, Pipelines), and Machine Learning (Playground, Experiments, Features, Models, Serving). The main area shows a SQL query in the 'New Query 2024-10-...' editor. The query is a SELECT statement filtering the 'system.access.audit' table for specific actions and client IDs, grouped by event_date and workspace_id. Below the query editor, a table of results is shown with columns: event_date, workspace_id, app, user_email, and username. The first row of data shows an event from 2024-10-08 at workspace 6051921418418893, with a null app, user_email '2d83bb93-3eb8-4f9c-ba13-c313836e8915', and a null username.

```
1 SELECT
2   event_date,
3   workspace_id,
4   request_params.request_object_id as app,
5   user_identity.email as user_email,
6   user_identity.subject_name as username
7 FROM
8   system.access.audit
9 WHERE
10  action_name IN ("workspaceInHouseOAuthClientAuthentication", "mintOAuthToken",
11                 "mintOAuthAuthorizationCode")
12 AND
13  request_params["client_id"] LIKE "2d83bb93-3eb8-4f9c-ba13-c313836e8915"
14 GROUP BY
15  event_date,
16  workspace_id,
17  app,
18  user_email,
19  username
```

	event_date	workspace_id	app	user_email	username
1	2024-10-08	6051921418418893	null	2d83bb93-3eb8-4f9c-ba13-c313836e8915	null

Pricing

Billed per hour based on app capacity

Standard capacity = up to 2 CPUs per app

- 0.5 DBU/hr
- = ~\$0.375/hr on AWS,
= ~\$0.475/hr on Azure (varies by region)
- May add different capacities in future
(eg, 4 CPU)

An app runs until you stop it

Appears on bill as:

AWS: All-Purpose Serverless Compute

Azure: Interactive Serverless Compute

System tables will have billing entries
for each app, with app id

+

New

📁

Workspace

🕒

Recents

📁

Catalog

🔄

Workflows

☁

Compute

SQL

📄

SQL Editor

📄

Queries

📊

Dashboards

🛠

Genie

🔔

Alerts

Compute

se compute

Job compute

SQL warehouses

Vector Search

Pools

GPU pools

Apps

ⓘ

...

🔍

Filter by keyword

✓

Owned by me

💬

Send feedback

Create app

Compute status ▾

Creator: nick.kramer@databricks.com ✕ ▾

Last updated ▾

3 matches

Reset filters

Name	Compute	Creator	Last update	App URL	
nick-data	paid  Active	nick.kramer...	4 days ago	Open app 	⋮
ntest	paid  Active	nick.kramer...	7 days ago	Open app 	⋮
shiny-nick	free  Stopped	nick.kramer...	4 days ago	Unavailable	⋮

Best Practices

Most Databricks best practices apply, but with a few extras:

Databricks

- Do your heavy data crunching in SQL warehouses, jobs, or model serving, not in apps.
- Use different workspaces to isolate dev, staging and production
- Use serverless egress controls to restrict outbound traffic

Identity

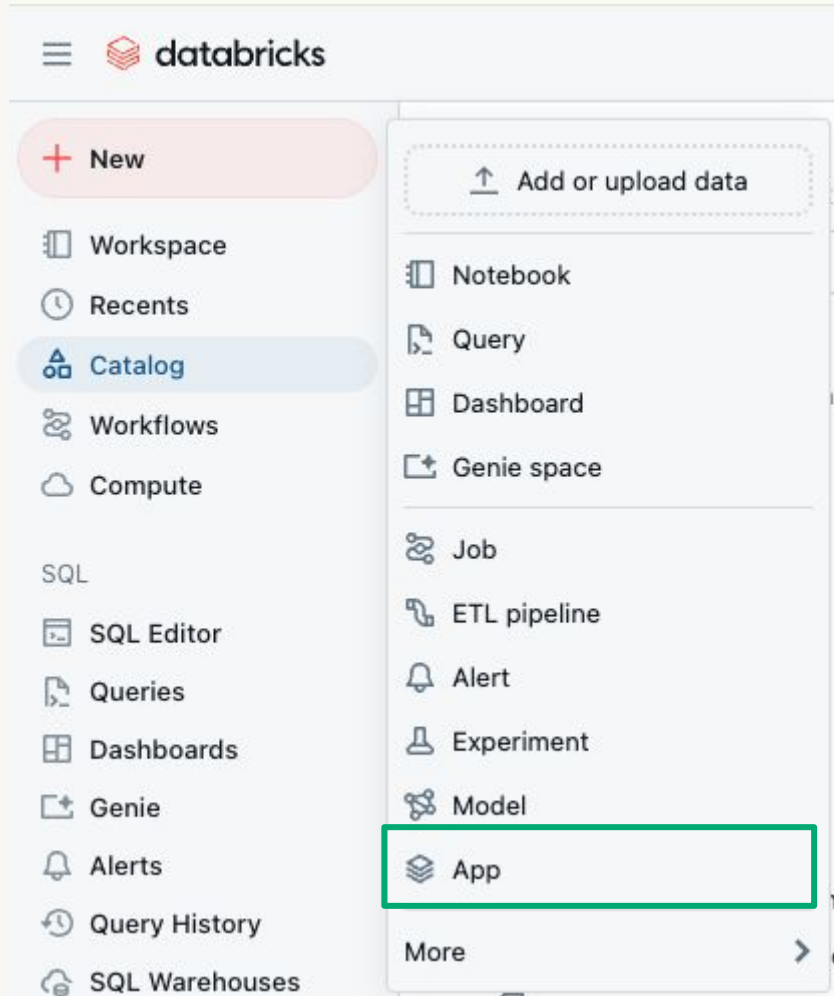
- Restrict the Can Manage app permission
- Follow the principle of least privilege when assigning permissions to the app's service principal
- Rotate the app's service principal credentials if the app creator leaves your organization
- Use secrets to store and reference your credentials
- Monitor audit logs for signs of suspicious activity

SDLC

- Use CI/CD to promote code & assets between different environments
- Parameterize your queries and be careful with SQL Injection
- Perform regular static application security testing (SAST) to find vulnerabilities in your code
- Perform regular static analysis to find vulnerabilities in your dependencies and patch them whenever necessary
- Follow the best practices relating to your chosen app framework (i.e. Gradio, Streamlit, Flask, Dash)



Get Started with Databricks Apps



To write your first app, go to **+ New** and click **Apps**.

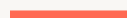
Follow the instructions on the screen. Make changes using your favorite source code editor, and deploy!

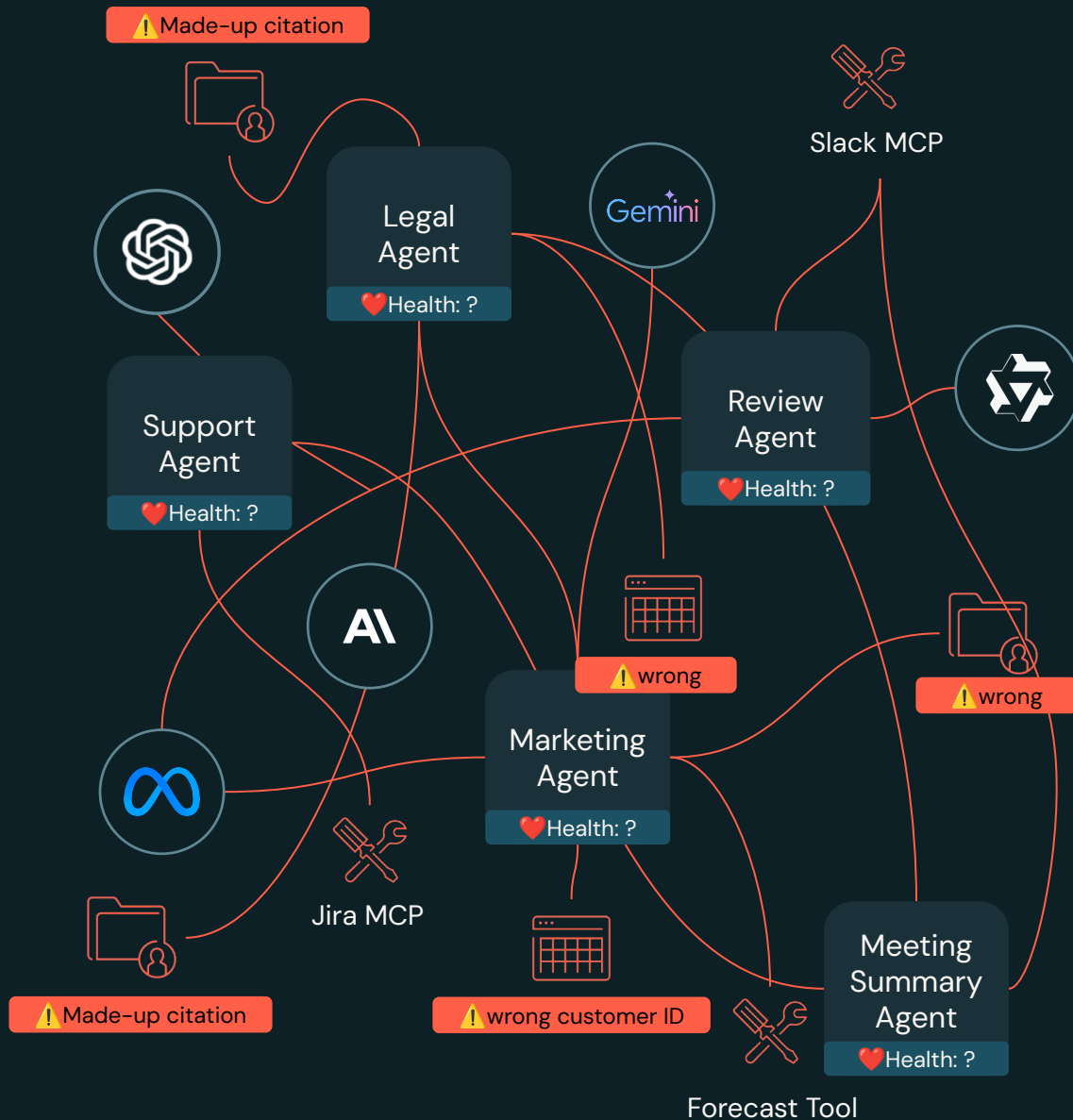
See the [documentation](#) (regional availability: [AWS](#), [Azure](#)) for more information on all the features.



Agent Bricks Deep Dive

Simplify the process of
building production agentic
solutions, brick by brick





There is **Agent Sprawl**

Low quality reasoning on enterprise data

No visibility or auditability

Too many AI vendors

No way to measure quality

Agent Bricks:

The control plane to
build and manage
enterprise agents



Contextual reasoning



Runs on any model or
framework natively



Evaluates & improves quality
over time



Unified catalog
and governance

Learned Semantics

Personalized table interactions

Tool & table popularity

Business Semantics

Metrics View
Agent Metadata

Custom Semantics

Extracted metadata from text
Parsed documents (images)



Agent Bricks Contextual Reasoning

Genie: state-of-the-art text2SQL generation

How does the total attributed revenue from AutoML users vary by date?

Analysis (click to view) automl_aggregate

This result is reviewed.

The total attributed revenue from AutoML users varies significantly by date, with some days showing much higher revenue than others. Examples include:

- 2021-02-15: 3.90
- 2021-03-19: 125.08
- 2021-05-28: 335.31
- 2021-06-01: 362.91
- 2021-06-03: 303.28



Contextual reasoning



Runs on any model or framework natively



Evaluates & improves quality over time



Unified catalog and governance

Natively serve all frontier AI models
one contract, any model, no lock-in



**Build custom agents with
any framework**



Across any cloud



Contextual reasoning



Runs on any model or
framework natively



Evaluates & improves quality
over time



Unified catalog
and governance

Tracing & Debugging

mlflow

Is same-day delivery becoming standard for electronics? and where will we have issues implementing this

ID: tr-a7d93dfa381b8f307eff4df29e81ad5a Latency: 2.67m Status: OK

Summary Details & Timeline Linked prompts

Search

Trace breakdown

0s 50.00s 1.67m 2.50m 3.33m

predict_stream 2.67m

examples 0.37s

agent-supplychainagent_1 3.95s

operational_docs 32.34s

agent-supplychainagent_2 15.86s

_query_genie_as_agent_2 15.60s

_concat_messages_array_2 0.20ms

ask_question_2 15.60s

start_conversation_2 0.14s

poll_for_result_2 15.46s

poll_result_2 15.46s

genie_timeline_2 15.46s

submitted_2 0.55s

filtering_context_2 0.56s

Inputs / Outputs Attributes Events Assessments

Inputs

request

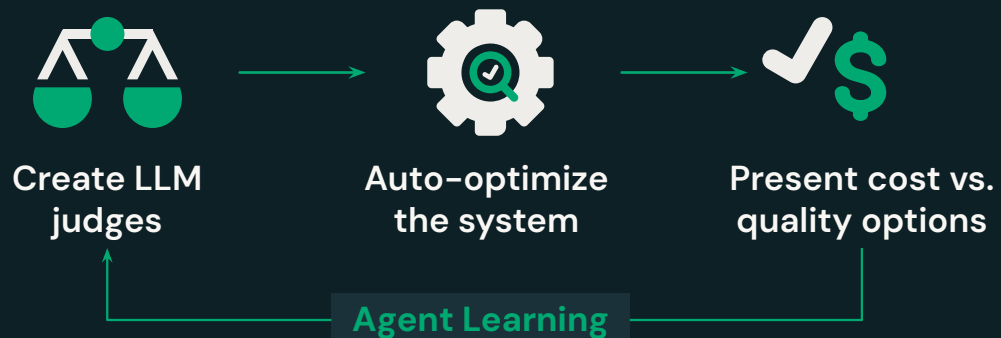
What are the current inventory levels for electronics products? Show me which electronics products have low stock levels, stockout risks, and any supplier performance issues that could impact same-day delivery capabilities. Focus on products that are frequently out of stock or have long lead times.

Outputs

response

	product_id	product_n...	category	available_...	stockout_r...	days_of_s...	lead_time_...
0	SKU_10525	Card Gaming Console	Electronics	0	HIGH	0	48
1	SKU_10525	Card Gaming Console	Electronics	0	HIGH	0	48

Automatic Improvements



Contextual reasoning



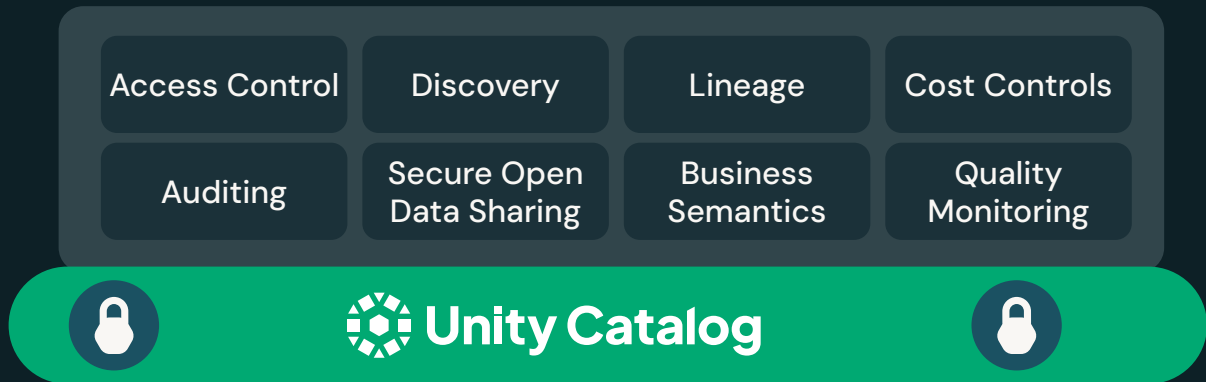
Runs on any model or framework natively



Evaluates & improves quality over time



Unified catalog and governance



Agents

MCP Servers

Models

Skills

Data

External Agents



Contextual reasoning



Runs on any model or framework natively



Evaluates & improves quality over time



Unified catalog and governance



Agent Bricks
Production AI agents



AI/BI
Agentic business intelligence



Custom Apps
Secure data and AI apps

And more...

Agent Platform

Contextual Reasoning Agents

Knowledge Assistant

Supervisor Agent

Documents Agent

AI Functions

Developer Platform

Agent Orchestration

Runtime 

Memory

Model Capacity 

AI Governance

Agent/Skill/MCP Registry

AI Gateway

Agent Observability

Managed OAuth Apps



